



A novel image encryption scheme based on an elliptic curve

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ABSTRACT

Elliptic curve cryptography (ECC) is capable of providing high security than to other cryptosystems with same key size. The aim of this paper is twofold. Firstly, we present new methods for the construction of substitution boxes (S-boxes), and the generation of pseudo random numbers (PRN) by using a total order on an elliptic curve (EC) over a prime field. A search method is used to efficiently generate an EC instead of the more traditional group law which is computationally expensive. The S-box generation technique uses the x -coordinates of the points of an ordered elliptic curve (OEC), while a generalization of the Frobenius map and n -norm are used on the points of an OEC to generate PRN. Secondly, a two phase image encryption system based on the newly developed S-box and PRN generation methods is proposed. In this security system, the plain-image is first diffused by masking it by the proposed PRN which is then confused by a proposed dynamic S-box. Rigorous analysis and comparison with some of the existing S-box and image encryption methods reveal that the proposed techniques are capable of generating cryptographically strong S-boxes, PRN with high entropy and optimal resistance against modern image cryptanalysis.

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1. Introduction

Security of confidential data has received much attention by cryptographers in the past few decades. Different types of information security techniques have been proposed by the researchers. Cryptography and steganography are the two main categories of the information security techniques [1]. The main idea of cryptographic techniques is to transform confidential data into an unreadable form by using key(s). In steganography, the secret data is secured from adversaries by embedding it into host data. Shannon [2] proved that for a strong security system it is necessary to generate confusion and diffusion in the data up to a certain level. There are different types of cryptosystems based on different mathematical structures. In many cryptographic techniques, an S-box is solely responsible for confusion and diffusion in the input data. Similarly, elliptic curves (ECs) are also used in developing secure cryptosystems. Miller first time introduced a cryptosystem based on ECs in [3] which is 20 percent faster than Diffie-Hellman protocol. Another fast and secure cryptosystem is proposed by Neal et al. in [4] based on the concept of a discrete logarithmic problem on ECs. Amara et al. [5] compared ECC with RSA and noticed that ECC has higher security than that of RSA.

The aim of this paper is to present novel and efficient algorithms for the construction of cryptographically strong S-boxes, pseudo random numbers (PRN) with high entropy, and security of confidential images based on elliptic curves over prime fields. The newly developed S-box technique uses the x -coordinates of points of a finite OEC followed by modulo operation 256. The proposed PRN generation scheme is based on a generalization of the Frobenius map, and n -norm over a finite OEC, while the proposed image cryptosystem is based on a dynamic S-box and PRN generated by the newly developed methods.

The rest of the paper is organized as follow: Section 2 contains some preliminaries. A survey of existing S-box generation techniques is given in Section 3. A new algorithm for the generation of S-boxes over finite OEC, their analysis and comparison are presented in Section 4. Section 5 contains the details and analysis of proposed PRN generation method. A short survey of existing image encryption techniques is given in Section 6. Section 7 contains a new image encryption system based on the proposed S-box and PRN generation techniques. Moreover, a detailed analysis and comparison of the newly developed image security system is presented in Section 7. Finally, conclusion and future directions are discussed in Section 8.

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2. Preliminaries

Let p be a prime and $\mathbb{F}_p$ denote the prime field of p elements. For any two non-negative integers $a, b \leq p$, the non-singular elliptic curve $E(p, a, b)$ over the field $\mathbb{F}_p$ is defined as

$$E(p, a, b) = \{(x, y) \in \mathbb{F}_p^2 \mid (y^2 = x^3 + ax + b) \pmod{p}\} \cup \{\infty\}, \quad (1)$$

provided the discriminant $(4a^3 + 27b^2) \pmod{p}$ of the EC is non-zero. We call p, a and b the parameters of the EC $E(p, a, b)$. Henceforth, we refer to an elliptic curve over a prime field by simply elliptic curve unless stated otherwise, the number of elements in a finite set A is denoted by $\#A$ and its i -th element is denoted by a_i . The number of points $\#E(p, a, b)$ on an EC $E(p, a, b)$ is of great importance. But, finding the exact value of $\#E(p, a, b)$ in general is not an easy task. However, a bound on the number of points on an EC $E(p, a, b)$ can be calculated using Hasse's Theorem [6] as

$$p + 1 - 2\sqrt{p} \leq \#E(p, a, b) \leq p + 1 + 2\sqrt{p}. \quad (2)$$

The surjective map $\pi(x, y) \mapsto (x^p, y^p) \pmod{p}$ from EC $E(p, a, b)$ to itself is called a *Frobenius map*.

For any positive real number $n \geq 1$, the n -norm is a function $\|\cdot\|_n$ from a space of vectors $\mathbf{v}$ of length m defined by $\|\mathbf{v}\|_n \mapsto (\sum_{i=1}^m |v_i|^n)^{1/n}$. Actually, for $n > 1$ the n -norm is a generalization of the Euclidean norm.

3. A brief introduction to S-box construction techniques

The National Institute of Standards and Technology (NIST) has adopted the Rijndael block cipher [7] as the Advanced Encryption Standard (AES). AES is one of the most preferred cryptosystems. Due to the significance of AES many cryptographers have explored the cryptographic properties of its S-box. Murphy et al. calculated an algebraic representation of the AES S-box in [8]. They also observed that AES can be represented by a system of quadratic equations. The non-linear part of AES is further investigated by Rosenthal in [9] by examining its polynomial representation. It is concluded that the AES S-box has low algebraic complexity because it has few non-zero terms in its linear polynomial, and hence AES might be vulnerable against algebraic attacks.

As an S-box plays a vital role in AES, therefore many researchers have suggested its improvements using various mathematical concepts including but not limited to algebraic and differential equations. Lui et al. [11] upgraded the AES S-box by swapping the orders of the transformations of the AES S-box. Cui et al. [12] deployed an affine function to enhance the complexity of the AES S-box against various algebraic attacks such as algebraic complexity. Gray codes have been used in [13] with the AES S-box to maximum the number of non-zero terms. Khan et al. [14,15] used group action, orbit of power function and affine mappings to generalise Gray S-box to construct multiple S-boxes. Hussain et al. [16] used Mobius transformations to construct new S-boxes. Guoping et al. [17] designed new S-boxes using logistic and the Baker maps. A new method using Baker and Chebyshev maps is presented in [18] by Guo et al. Wang et al. [19] constructed new S-boxes using artificial neural networks and a chaotic map. Jung et al. [20] explored a relationship between the non-linearity of S-box and hyper-elliptic curves. Hayat et al. [21] proposed a construction of S-boxes based on finite EC.

4. The proposed S-box generation algorithm

The proposed S-box generation algorithm uses a total order on an EC which is defined in Section 4.1.

4.1. A total order on an EC

For any prime p and any two non-negative integers $a, b \leq p$, let $E(p, a, b)$ be the elliptic curve $y^2 = x^3 + ax + b$. A *priority total order* " $<$ " [10] on the set $E(p, a, b)$ is defined as

$$(x, y) < (x', y') \Leftrightarrow \begin{cases} \min\{x, y\} < \min\{x', y'\}; \\ \min\{x, y\} = \min\{x', y'\} \text{ and } y < y'; \text{ or} \\ \min\{x, y\} = \min\{x', y'\}, y = y' \text{ and} \\ x < x' \end{cases} \quad (3)$$

for all $(x, y), (x', y') \in E(p, a, b)$. Henceforth, we assume the priority total order $<$ on $E(p, a, b)$ and $E(p, a, b)^<$ denote the set containing all elements of $E(p, a, b) \setminus \{\infty\}$ arranged in ascending order according to the definition of $<$. We call $E(p, a, b)^<$ an *ordered elliptic curve* (OEC). The element at i -th position $e(p, a, b)_i^<$ in the set $E(p, a, b)^<$ is denoted by (x_i, y_i) . The aim of this total order is to diffuse the x -coordinates of points of $E(p, a, b)$ in a typical way. Further, we define a multi-set $E'(p, a, b, x)^<$ consisting of x -coordinate of all points of $E(p, a, b)^<$ such that $e'(p, a, b, x)_i^< = x_i$ i.e., the order is preserved. Notice that

$$\#E(p, a, b)^< = 2 \times \#E(p, a, b, x)^<. \quad (4)$$

This is because of the fact that for each value of x , there are two values of y satisfying the equation $y^2 = x^3 + ax + b$.

4.1.1. The algorithm

The proposed S-box generation algorithm consists of five main steps which are given below

Step 1. Select a prime p and any two non-negative integers a and b such that $a, b \leq p - 1$. The value of p is selected so that the OEC $E(p, a, b, x)^<$ has at least 256 distinct elements. This restriction is necessary, since an S-box on the Galois field $GF(2^8)$ has 256 distinct elements. Thus from (2) and (4), the selected value of p for the proposed algorithm must satisfy

$$p + 2 - 2\sqrt{p} \geq 2 \times 256. \quad (5)$$

Step 2. Generate the ordered elliptic curve $E(p, a, b)^<$ by using brute force search method, and the priority total order.

Step 3. Now, apply function mod 256 on the elements of $E'(p, a, b, x)^<$ to get another multi-set $E''(p, a, b, x, 256)^<$ such that $e''(p, a, b, x, 256)_i^< = e'(p, a, b, x)_i^< \pmod{256}$. This operation is used to restrict the values of the elements of $E'(p, a, b, x)^<$ in the range $0 - 255$. Of course, $\#E''(p, a, b, x, 256)^< = \#E'(p, a, b, x)^<$.

Step 4. Next, select a positive integer ℓ such that

$$\#E''(p, a, b, x, 256)^< - \ell \geq 256. \quad (6)$$

The constraint on the selection of ℓ is imposed so that $\#E''(p, a, b, x, 256)^<$ has at least 256 elements after the $(\ell - 1)$ -th element.

Step 5. Finally, an S-box $S_{p,\ell}^{a,b}$ is generated by selecting the first 256 distinct integers from ℓ -th element of the multi-set $E''(p, a, b, x, 256)^<$. Note that if there are not 256 distinct integers following the ℓ -th element in $E''(p, a, b, x, 256)^<$, then no S-box is found by the proposed method.

It is clear from the step 5 that the newly constructed S-box is bijective. The proposed algorithm is implemented on several ECs for the generation of S-boxes. For example, the S-box $S_{3917,1}^{16,3211}$ generated by the proposed method is presented in 16×16 standard format in Table 1.

4.2. Analysis and comparison

Different standard security performance tests are applied on randomly generated different S-boxes $S_{2557,4}^{843,669}$, $S_{2266,2155}^{1195,2950}$, $S_{3413,1}^{3613,1}$, $S_{29,461}^{16,3211}$, $S_{29,461}^{2850,2230}$, $S_{3922,3717}^{3926,2411}$, $S_{3917,1200}^{16,3211}$, $S_{3917,1}^{2850,2230}$, $S_{4289,1}^{5897,1}$, $S_{5903,300}^{6247,1}$.

Table 1
The S-box $S_{3917,1}^{16,3211}$ generated by the proposed algorithm.

56	236	135	231	21	171	5	145	92	124	141	46	6	241	16	187
89	214	36	244	150	53	247	107	147	238	127	42	196	249	11	125
83	126	27	220	54	157	181	130	94	161	155	123	185	93	112	188
242	211	14	129	144	232	173	96	23	90	128	111	167	25	170	162
146	239	223	30	31	32	57	174	132	217	44	63	50	137	74	218
51	75	28	15	85	52	62	136	201	253	86	104	215	140	156	153
213	19	159	120	230	114	154	131	70	190	184	164	4	2	110	233
29	197	229	198	200	192	60	9	116	68	166	193	199	26	95	240
88	37	182	0	176	149	105	221	204	49	40	175	179	67	20	117
216	69	224	61	72	163	100	234	98	248	81	118	59	13	252	58
66	8	235	22	139	121	17	134	122	91	152	254	237	103	18	7
1	80	255	222	35	165	169	71	102	191	148	10	3	160	202	43
226	245	208	113	178	47	243	24	246	172	142	189	48	39	106	186
183	115	64	45	158	219	108	177	206	55	109	227	79	251	225	250
212	78	12	65	203	143	210	209	38	34	82	195	77	33	76	151

Table 2
Experimental results of the newly generated S-boxes and some of the existing S-boxes.

S-boxes	NL	LAP	SAC		BIC		DAP	AC
			min	max	min	max		
$S_{843,669}^{2557,4}$	104	0.1406	0.4063	0.5938	0.4570	0.5293	0.0391	254
$S_{2266,2155}^{3413,1}$	104	0.1328	0.4063	0.5938	0.4668	0.5430	0.0391	253
$S_{1195,2950}^{3613,1}$	104	0.1328	0.4063	0.5938	0.4629	0.5254	0.0391	255
$S_{29,461}^{3917,1200}$	104	0.1484	0.4219	0.6094	0.4629	0.5430	0.0469	255
$S_{16,3211}^{3917,1}$	106	0.1484	0.4375	0.5938	0.4648	0.5449	0.0391	255
$S_{29,461}^{3917,1}$	106	0.1484	0.4375	0.5781	0.4648	0.5352	0.0391	254
$S_{2850,2230}^{4289,1}$	104	0.1484	0.4063	0.5781	0.4766	0.5488	0.0391	253
$S_{3922,3717}^{5897,1}$	104	0.1328	0.4219	0.5781	0.4766	0.5391	0.0391	252
$S_{3926,2411}^{5903,300}$	104	0.1328	0.4063	0.6094	0.5430	0.4727	0.0391	254
$S_{2073,3708}^{6247,1}$	104	0.1328	0.375	0.5938	0.4492	0.5332	0.0391	255
$S_{2311,5678}^{6961,1}$	104	0.1250	0.3906	0.5781	0.4707	0.5352	0.0391	255
$S_{3123,8707}^{9397,1}$	104	0.1406	0.4219	0.5781	0.4746	0.5371	0.0391	253
$S_{6358,2836}^{9551,1}$	104	0.1406	0.4063	0.6095	0.4727	0.5254	0.0391	255
Ref. [22]	108	0.156	0.406	0.502	0.47	0.503	0.046	255
Ref. [23]	98	0.0352	0.4453	0.5781	0.4922	0.5156	0.046	255
Ref. [17]	103	0.0352	0.4414	0.5703	0.4961	0.5039	0.0391	255
Ref. [18]	102	0.078	0.375	0.6094	0.4707	0.5215	0.0391	254
Ref. [24]	104	0.109	0.39	0.593	0.454	0.499	0.0469	255
Ref. [19]	106	0.0469	0.4375	0.5938	0.4648	0.5313	0.0391	251
Ref. [16]	100	0.125	0.493	0.593	0.0137	0.476	0.0391	255

$S_{6961,1}^{2311,5678}$, $S_{9397,1}^{3123,8707}$, and $S_{9551,1}^{6358,2836}$ by the proposed algorithm in this section. Furthermore, the experimental results are compared with some of the known S-boxes presented in [16–19,22–24] constructed by different mathematical approaches.

4.2.1. Non-linearity (NL)

The concept of non-linearity is used to measure the ability of an S-box to create confusion [25]. For a given S-box $S: GF(2^8) \rightarrow GF(2^8)$, the NL $\delta(S)$ of S is defined to be its distance to the affine functions over $GF(2^8)$

$$\delta(S) = \min_{\alpha, w, \beta} \{x \in GF(2^8) \mid \alpha \cdot S(x) \neq \beta \cdot x \oplus w\}, \quad (7)$$

where $\alpha \in GF(2^8)$, $w \in GF(2)$, $\beta \in GF(2^8) \setminus \{0\}$ and “ $\cdot$ ” denotes the dot product over $GF(2)$.

The ideal value of non-linearity that an S-box may attain is 120. It is known fact that if $\delta(S)$ is 120 then the results of other cryptographic tests on S may not lie in the desire range [25]. Therefore S-boxes with optimal NL and satisfying other security performance tests are of great importance. We computed the non-linearity of the newly designed S-boxes by the proposed algorithm and some other existing S-boxes. The experimental results of NL test are listed in Table 2. The maximum, average and minimum values of NL of the proposed S-boxes are 106, 104.308 and 104, respectively. This implies that the newly developed S-boxes have good resistance against linear attacks. From Table 2, it is easy to

notice that the NL of proposed S-boxes is greater than that of the S-boxes presented in [16–18,23,24]. Hence, the proposed S-box generation technique can generate good S-boxes when compared to some of the other S-box generation techniques.

4.2.2. Linear approximation probability (LAP)

Mitsuru et al. [26] introduced this test to calculate the probability of getting a linear approximation of a given S-box. The LAP of an S-box is dependent on the coincidence of bits of plain-text and cipher-text. The mathematical formula of LAP is given in Equation (8)

$$N(\alpha, \beta) = \#\{x \in GF(2^8) \mid \alpha \cdot x = \beta \cdot S(x)\} - 2^{n-1},$$

$$LAP(S) = \frac{1}{2^n} \left\{ \max_{\alpha, \beta} |N(\alpha, \beta)| \right\}, \quad (8)$$

where $\alpha \in GF(2^8)$, $\beta \in GF(2^8) \setminus \{0\}$ and “ $\cdot$ ” denotes the dot product over $GF(2)$.

An S-box is said to be highly resistive against linear attacks if it has low LAP. The experimental results of LAP are listed in Table 2.

The maximum, average and minimum values of the proposed S-boxes are 0.1484, 0.1388 and 0.1250, respectively, which are closer to the optimal value. Hence, the proposed S-boxes have good resistance to linear attacks.

Table 3SAC dependence matrix of the newly generated S-box $S_{3917,1}^{16,1088}$.

0.4531	0.5313	0.4688	0.4844	0.4844	0.4375	0.4531
0.5156	0.4844	0.4844	0.5313	0.5156	0.5469	0.4688
0.4844	0.5469	0.4375	0.5	0.5625	0.4375	0.4844
0.5625	0.4844	0.5	0.4531	0.5156	0.5	0.5625
0.5625	0.4531	0.5781	0.4844	0.5156	0.5	0.4688
0.4688	0.5313	0.4844	0.5625	0.5	0.5	0.5313
0.5469	0.5313	0.5313	0.4531	0.5156	0.4688	0.5469
0.4688	0.5469	0.4688	0.4688	0.4375	0.4688	0.5

Table 4BIC matrix of the newly generated S-box $S_{3917,1}^{16,1088}$.

–	0.4941	0.5098	0.5078	0.4902	0.498	0.5176	0.5098
0.4941	–	0.4941	0.502	0.4648	0.5059	0.5234	0.5
0.5098	0.4941	–	0.5215	0.5078	0.5117	0.5117	0.4941
0.5078	0.502	0.5215	–	0.4785	0.4746	0.5039	0.5449
0.4902	0.4648	0.5078	0.4785	–	0.498	0.498	0.5215
0.498	0.5059	0.5117	0.4746	0.498	–	0.4824	0.5117
0.5176	0.5234	0.5117	0.5039	0.498	0.4824	–	0.5137
0.5098	0.5	0.4941	0.5449	0.5215	0.5117	0.5137	–

4.2.3. Strict avalanche criterion (SAC)

This criterion is introduced by Webster et al. [27] by using a combination of the avalanche effect and completeness. The test calculates the probability of change in a cipher bit when one input bit is inverted. The SAC of an S-box is computed using an 8×8 dependence matrix $M = [m(i, j)]_{i=1, j=1}^{8, 8}$ such that

$$m(i, j) = \left\{ \frac{1}{2^8} \left[w(S_i(x + \alpha_j) \oplus S_i(x)) \right] \mid \alpha_j \in GF(2^8), w(\alpha_j) = 1 \text{ and } 1 \leq i, j \leq 8 \right\}, \quad (9)$$

where $w(\alpha_j)$ is the number of non-zero bits in α_j .

The SAC is satisfied if all entries of the dependence matrix are near to 0.5. The SAC dependence matrix of the proposed S-box $S_{3917,1}^{16,3211}$ is presented in Table 3. Furthermore, we implemented the SAC test on other proposed and existing S-boxes, and the results are listed in Table 2. The average of the SAC maximum and minimum of the proposed S-boxes are 0.5914 and 0.4111, respectively. Thus, the S-boxes constructed by the proposed method satisfy the SAC test. Furthermore, the SAC results of newly developed S-boxes are comparable with the SAC results of the existing listed S-boxes.

4.2.4. Bit independence criterion (BIC)

The BIC test is also introduced by Webster et al. [27] to observe how independent the pair of output bits are when one input bit is being complemented. In this test one bit is complemented and its effect on the output bits is observed by calculating the correlation coefficient. An S-box is said to be satisfying the BIC test if its BIC value is near to 0.5. We implemented the BIC test on the proposed and some of the existing S-boxes. The BIC matrix of the S-box $S_{3917,1}^{16,3211}$ is given in Table 4, while the BIC minimum and maximum of other S-boxes are listed in Table 2. The average of the BIC maxima and minima of the proposed S-boxes are 0.5317 and 0.4725, respectively, which are near to 0.5. Furthermore, it is clear from Table 2 that the BIC values of the newly generated S-boxes are better than that of the S-boxes in [16,24]. Hence, the S-boxes generated by the proposed method efficiently satisfy BIC when compared to the S-boxes in [16,24].

4.2.5. Differential approximation probability (DAP)

Shamir et al. [28] proposed this test to calculate the probabilistic effect of a particular difference in the input bit on the difference of the resultant output bits. The mathematical formula used to calculate the DAP of an S-box S is

$$DAP(S) = \max_{\Delta x, \Delta y} \left\{ \# \left\{ x \in GF(2^8) \mid S(x + \Delta x) - S(x) = \Delta y \right\} \right\}, \quad (10)$$

where $\Delta x, \Delta y \in GF(2^8)$.

An S-box resists the differential attacks efficiently if it has low DAP. The DAP matrix of the newly generated S-box $S_{3917,1}^{16,3211}$ is given in Table 5. Similarly, the DAP of the other S-boxes used in this paper is listed in Table 2. The DAP value of some of the proposed S-boxes is 0.0397 which is less than the DAP value of the S-boxes in [22–24]. This implies that the S-boxes generated by the proposed method can resist differential attacks more effectively when compared to some of the existing S-boxes.

4.2.6. Algebraic complexity (AC)

Niederreiter et al. [29] introduced the concept of a linear polynomial for an S-box. The number of non-zero terms in the linear polynomials of an S-box measures its algebraic complexity. A cryptographically secure S-box must have high AC. The maximum value of AC that an S-box can have is 255. In Table 6, the coefficients of polynomial corresponding to the S-box $S_{3917,1}^{16,3211}$ are presented, while the AC of the other newly generated S-boxes and some of the existing S-boxes are listed in Table 2. It is evident from the experimental results that the newly generated S-box can efficiently resist algebraic attacks, since they have AC very close to the optimal value.

5. Pseudo random numbers (PRN) generation scheme

Random numbers and their generation play a vital role in many cryptographic applications such as encryption, key generation, internet gambling and masking protocols. Several random number generation methods are proposed based on different mathematical structures. ECs are also used for the generation of PRN [30–33]. Usually, the EC based PRN generation methods use the group law for the generation of points on an EC which requires high computation effort. In this section, we propose an efficient method to generate PRN over an OEC based on a search method, a generalization of the Frobenius map, and an n -norm.

Let p be a prime and q be a positive integer. A generalized Frobenius map γ_q from any EC $E(p, a, b) \setminus \{\infty\}$ to $\mathbb{F}_p^2$ is defined as $(x, y) \mapsto (x^q, y^q) \pmod{p}$. We denote the image of i -th element (x_i, y_i) of an EC under the generalized Frobenius map (GFM) by $(x_{i'}, y_{i'})$. Clearly, the generalized Frobenius map γ_q becomes identical to the Frobenius map π when $q = p$.

5.1. The algorithm

The proposed algorithm calculates a non-negative random integer corresponding to each point of an OEC by using the following steps

Step 1. Select a large prime p and any two non-negative integers $a, b \leq p - 1$.

Step 2. Generate the ordered elliptic curve $E(p, a, b)^<$ by using a brute force search instead of the group law to reduce the time complexity of the proposed algorithm.

Step 3. Fix a positive integer q and apply the generalized Frobenius map on each point (x_i, y_i) of $E(p, a, b)^<$ i.e., $\gamma_q(x_i, y_i) = (x_i^q, y_i^q) \pmod{p}$, where $i \leq \#E(p, a, b)^<$. The aim of the application

Table 5DAP matrix of the newly generated S-box $S_{3917,1}^{16,3211}$.

0.0234	0.0234	0.0234	0.0234	0.0313	0.0234	0.0234	0.0234	0.0313	0.0313	0.0313	0.0313	0.0234	0.0313	0.0234	0.0234
0.0313	0.0391	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0313	0.0313	0.0234	0.0234	0.0234	0.0234
0.0313	0.0234	0.0234	0.0313	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0391	0.0313	0.0234	0.0234	0.0234
0.0234	0.0234	0.0234	0.0313	0.0234	0.0313	0.0234	0.0234	0.0234	0.0313	0.0313	0.0234	0.0313	0.0313	0.0234	0.0234
0.0234	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0391	0.0234	0.0234	0.0313	0.0234
0.0234	0.0313	0.0234	0.0313	0.0234	0.0234	0.0313	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234
0.0313	0.0234	0.0313	0.0234	0.0313	0.0234	0.0234	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234
0.0313	0.0234	0.0234	0.0234	0.0234	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0313	0.0234
0.0234	0.0313	0.0234	0.0313	0.0313	0.0391	0.0234	0.0234	0.0234	0.0313	0.0156	0.0234	0.0234	0.0313	0.0234	0.0234
0.0234	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0313	0.0234	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234
0.0234	0.0234	0.0234	0.0313	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0313	0.0313
0.0234	0.0234	0.0234	0.0234	0.0313	0.0234	0.0313	0.0234	0.0313	0.0234	0.0313	0.0313	0.0234	0.0313	0.0313	0.0234
0.0313	0.0313	0.0234	0.0234	0.0234	0.0234	0.0234	0.0234	0.0313	0.0391	0.0234	0.0234	0.0313	0.0313	0.0313	0.0234
0.0391	0.0234	0.0234	0.0313	0.0313	0.0234	0.0313	0.0234	0.0391	0.0313	0.0234	0.0313	0.0234	0.0313	0.0234	0.0391
0.0313	0.0234	0.0234	0.0313	0.0234	0.0313	0.0234	0.0234	0.0234	0.0313	0.0234	0.0313	0.0313	0.0234	0.0234	0.0313
0.0313	0.0234	0.0313	0.0156	0.0313	0.0234	0.0234	0.0313	0.0234	0.0234	0.0313	0.0234	0.0313	0.0234	0.0313	0

Table 6Coefficients of the linear polynomial of the newly generated S-box $S_{3917,1}^{16,3211}$.

0	69	238	77	47	166	169	11	47	220	72	71	106	186	164	143
120	198	137	159	220	135	20	153	246	65	86	159	182	167	234	187
17	15	123	247	103	140	84	142	174	82	47	170	47	91	65	63
176	191	45	233	94	110	132	99	49	46	95	120	114	125	215	176
145	230	189	175	118	184	203	155	143	233	23	185	155	169	237	27
90	228	76	241	40	141	117	206	65	163	185	173	28	148	218	129
150	251	115	238	239	61	156	26	138	178	210	157	167	162	195	90
25	34	82	154	139	210	195	101	27	122	77	95	254	140	18	214
188	158	122	38	2	31	130	161	38	164	27	222	233	140	137	47
182	123	49	246	40	47	250	241	171	62	158	179	33	62	124	180
160	105	70	147	48	29	162	60	252	37	249	240	250	12	181	190
208	97	67	242	178	90	22	190	163	161	156	150	57	203	162	241
34	63	69	251	57	129	62	25	56	48	244	113	4	215	214	130
117	11	114	217	239	161	150	210	108	220	205	42	151	216	197	234
206	54	119	152	158	184	237	100	127	8	29	175	63	35	57	207
48	100	216	203	167	7	147	225	132	193	100	13	144	135	31	56

of GFM is to project the points of OEC and create a key q for the proposed PRN generation algorithm which will be important for key based cryptography.

Step 4. Next, select a real number $n \geq 1$ and apply the n -norm on the result of Step 3 to get a real number i.e., $\|\gamma_q(x_i, y_i)\|_n$.

Step 5. Finally use the nearest integer function to get an integer $\lfloor \|\gamma_q(x_i, y_i)\|_n \rfloor$ from the result of Step 4.

Let $\mathcal{R}(p, a, b, q, n)$ be the multi-set obtained by the application of Steps 1 – 5 on each element of $E(p, a, b)^<$ whose i -th element is defined by

$$r(p, a, b, q, n)_i = \lfloor \|\gamma_q(x_i, y_i)\|_n \rfloor = \left\lfloor \sqrt[n]{x_i^n + y_i^n} \right\rfloor, \quad (11)$$

where $\gamma_q(x_i, y_i) = (x_i^q, y_i^q)$. The set $\mathcal{R}(p, a, b, q, n)$ is the required set of random numbers obtained by the OEC $E(p, a, b)^<$. Clearly, $\#\mathcal{R}(p, a, b, q, n) = \#E(p, a, b)^<$.

We implemented the proposed algorithm for different randomly selected parameters to generate different random number multi-sets $\mathcal{R}(p, a, b, q, n)$. In Fig. 1, the first 200 elements of some random multi-sets $\mathcal{R}(p, a, b, q, n)$ are shown. The randomness in the points is clear from Fig. 1. Moreover, it is easy to observe from Fig. 1 that the different values of parameters p, a, b, q and n generate totally different random numbers.

5.1.1. Entropy test

The entropy test is one of the most important and efficient tests to correctly measure the extent of randomness in a data set. Given a data set X with its range set Ω_X and the probability $P(x_i)$ of each of its element x_i , the entropy $H(X)$ of X is calculated as

$$H(X) = - \sum_{i=1}^{\#\Omega_X} P(x_i) \log_2(P(x_i)). \quad (12)$$

Table 7

Entropy test of different random number multi-sets obtained by the proposed scheme.

p, a, b, q, n	$\#\mathcal{R}$	$H(\mathcal{R})$	$\log_2(\#\Omega_{\mathcal{R}})$
80021,97,79241,7,5	79703	15.3110	16.2823
90001,2216,70897,11,3	89544	15.5642	16.4503
97171,66789,45,3,3	97585	15.6826	16.5743
100003,99611,27077,2,2	100170	15.1688	16.6120

A data is said to be highly random if its entropy is high. The upper bound of entropy is proved in [34] which is

$$H(X) \leq \log_2(\#\Omega_X). \quad (13)$$

We implemented the entropy test on the random numbers $\mathcal{R}(p, a, b, q, n)$ obtained by the variations of the parameters of $\mathcal{R}$. The results of these experiments are listed in Table 7. Notice that the value of entropy $H(\mathcal{R})$ of different random number sets are close to their upper bounds $\log_2(\#\Omega_{\mathcal{R}})$. This directly implies that the proposed algorithm is capable of generating highly random numbers based on elliptic curves.

6. A brief introduction to image encryption techniques

Security of confidential images has become one of the hot areas in cryptography. The traditional security systems, particularly, the cryptosystems relying on only a single S-box are not good enough to ensure the security of images [54]. This is because of the different nature of images compared to text in terms of high correlation among adjacent pixels, high data capacity and hence high redundancy. It is therefore necessary to develop some new cryptosystems which can efficiently resist the image security attacks. Many

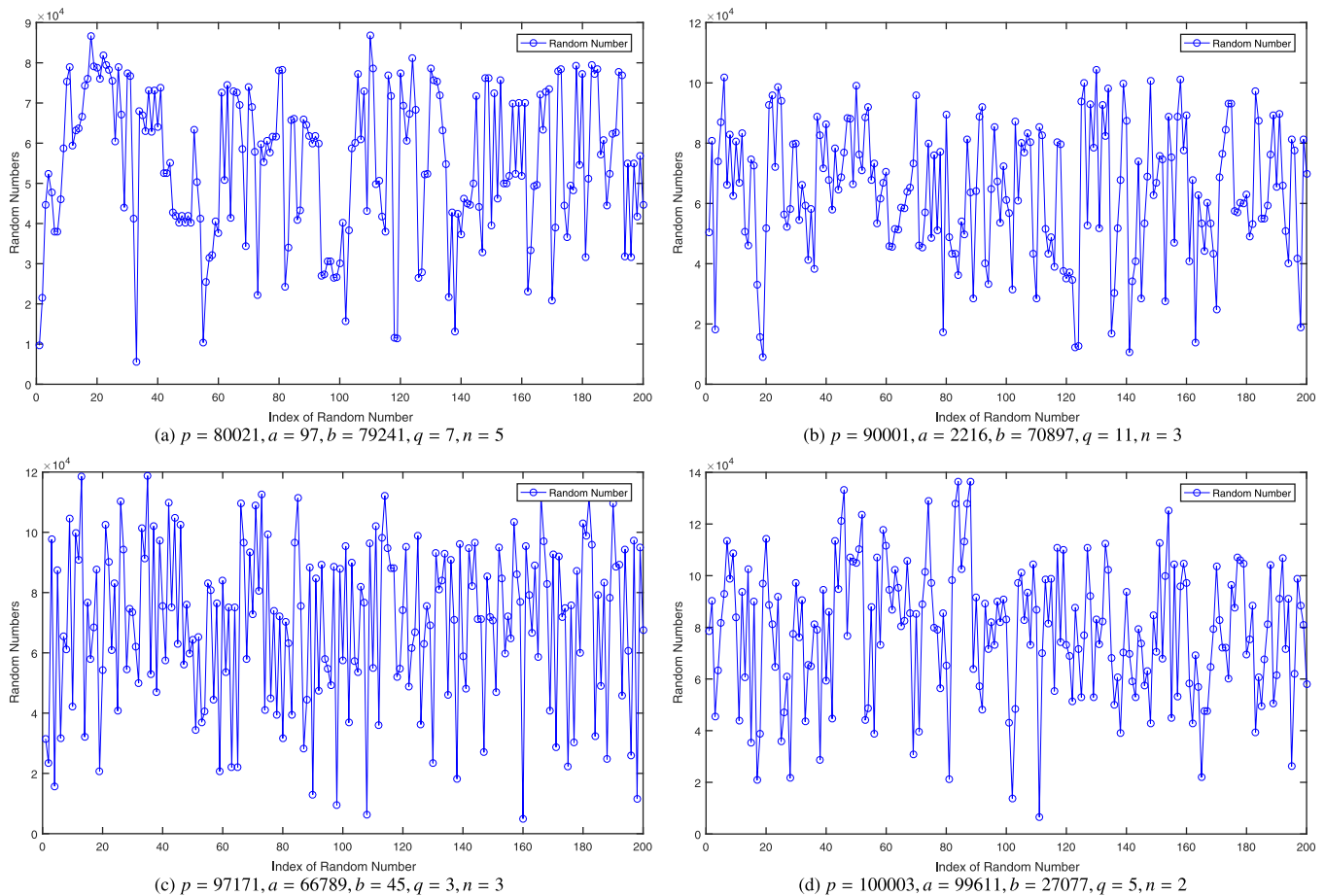


Fig. 1. The first 200 random numbers of different random number sets generated by the proposed algorithm.

researchers have proposed different image cryptosystems. Most of the time these cryptosystems can be divided into three categories:

(a) Sometimes multiple S-boxes are used instead of a single S-box for diffusion and confusion of secret images e.g., see [35] and references therein. Usually, with the increase in the number of multiple S-boxes not only the security but also the computation time increases. Thus, in this area the interest lies in reducing the number of S-boxes to achieve the desired security.

(b) In the second category, PRN are used for the security of images. These PRN are generated by different methods. In some cryptosystems, it is necessary to send all PRN as key for decryption. This increases the security of images but the drawback of such techniques is the increase in the data size. However, some cryptosystems require a few number of parameters to generate the random numbers. Such techniques include chaos and elliptic curve based random number generation methods. In chaos based random number generation, the interest lies in the selection of a highly chaotic system. Different types of chaotic maps such as Arnold's cat map, Lorenz system, Baker's map, and Chen chaotic attractor are used for the generation of random numbers. However, in the case of elliptic curve based random sequence, the main focus is on the method of generation of points on the EC. Usually, the group law on the EC is used for this purpose. The cryptographic efficiency of random numbers is highly dependent on the period of the underlying systems. That is, a sequence of random numbers is cryptographically strong if it has high period. However, chaotic sequences can have short periods depending on the nature of computational accuracy. Furthermore, in general the EC cryptography has high security compared to the chaos based cryptography [36]. This is be-

cause of the EC discrete logarithm problem is NP hard [37,38]. For more details on cryptography and random numbers the readers are referred to [33,36,39–41,43,51,52].

(c) The third category of image cryptosystems contains hybrid security systems based on S-boxes and PRN. Once again chaotic maps and elliptic curves are used in these techniques for the generation of random numbers. The number of S-boxes used in these techniques varies. Of course, the time complexity of a hybrid image encryption technique is directly proportional to the number of S-boxes it uses. Furthermore, some techniques use static S-boxes while others use dynamic S-boxes, see [30,45,46,49,53]. With some calculations it can be shown that the techniques using dynamic S-boxes are more secure than those using static S-boxes, since all images are not of the same nature [47]. However, to the best of our knowledge, until now no EC cryptosystem has been developed based on the concept of dynamic S-boxes.

7. The proposed image cryptosystem

The proposed encryption system is a two phase technique. In the first phase, the plain-image is masked by using the random numbers generated by the proposed algorithm discussed in Section 5.1. The aim of the masking phase is to diffuse the plain-image, particularly, the binary image. Note that the proposed random number generation technique is based on a search method to find all points on a given OEC instead of the group law. Hence, the masking phase requires less time than other EC techniques in which the masking is performed based on the group law. Now, to have high security against cryptanalysis it is necessary to generate confusion in the plain-image. Thus, in the second phase, confusion

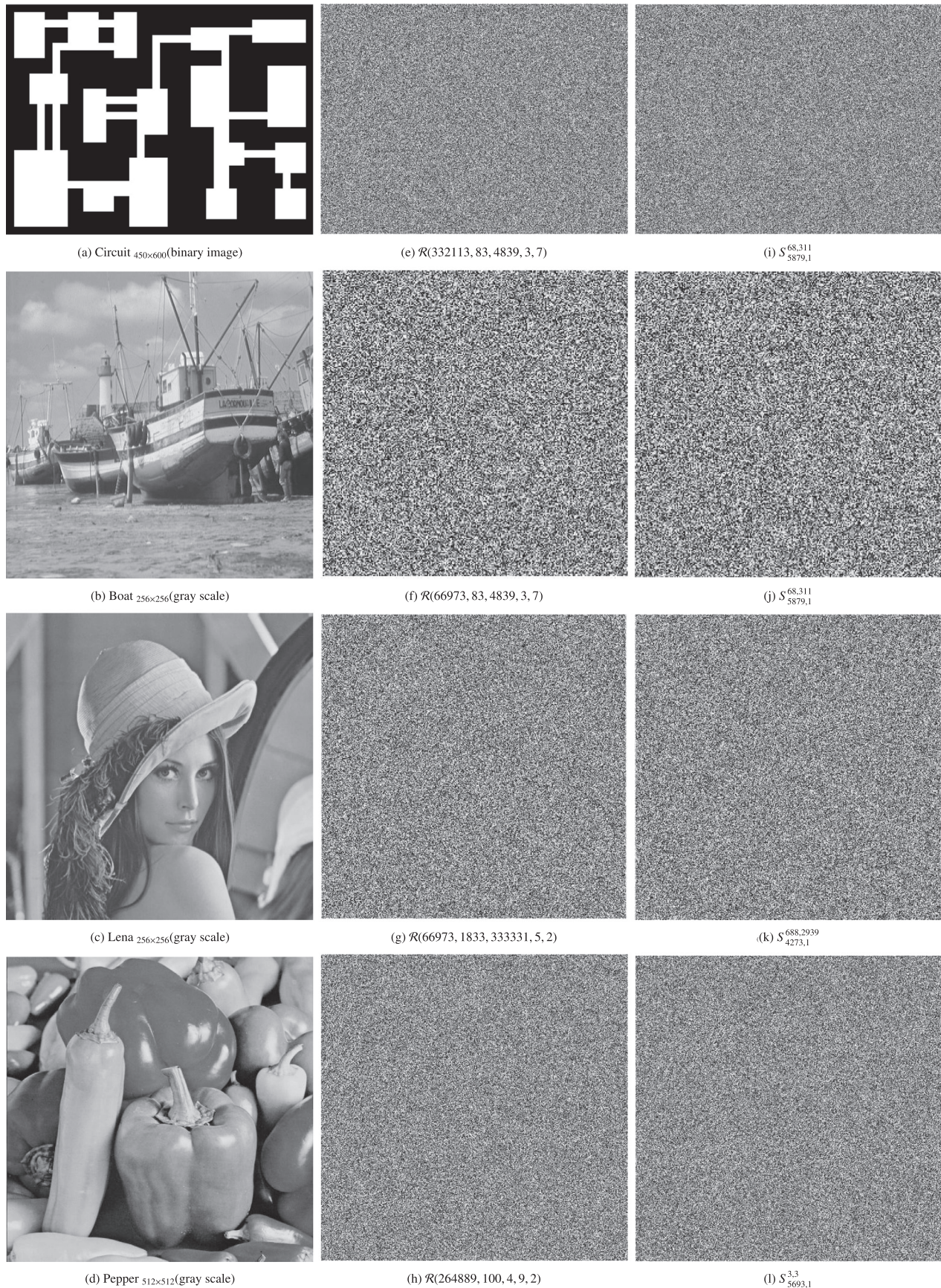


Fig. 2. (a-d) plain-images, (e-h) masked images and (i-l) encrypted images by the proposed method.

is created in the masked image by using a dynamic S-box generated by the proposed algorithm as described in Section 4.1.1.

Henceforth, a plain-image I of width W and height H is denoted by $I_{W \times H}$, the sum of all pixel intensities of I is denoted by ΣI , and the images obtained after masking and encryption of I are denoted by I_M and I_C , respectively.

7.1. The encryption process

Let $I_{W \times H}$ be the given plain-image. The main steps of the proposed encryption system are discussed below.

7.1.1. The masking phase

Step 1. Calculate the sum ΣI .

Step 2. Select a large prime p satisfying

$$p + 2 - 2\sqrt{p} \geq W \times H. \quad (14)$$

The constraint on p is necessary, since the proposed masking of the image I requires $W \times H$ points on the OEC.

Step 3. Select two non-negative real numbers k_1 and k_2 , and calculate the parameters a and b for the OEC to be used for masking by

$$a = \left(k_1 + \sum I\right) \pmod{p}, \quad (15)$$

$$b = \left(k_2 + \sum I\right) \pmod{p}. \quad (16)$$

Note that the parameters of the OEC are linked with the plain-image in Equations (15) and (16) so that the generated OEC depends on the input image.

Step 4. Now select a natural number q and a real number $n \geq 1$, and generate the multi-set $\mathcal{R}(p, a, b, q, n)$ of random numbers based on the method explained in Section 5.1.

Step 5. Finally, mask each pixel of the plain-image I to create diffusion by

$$I_M(i, j) = (I(i, j) + r(p, q, a, b, n)_{H \times (i-1) + j}) \pmod{256} \quad (17)$$

$$= (I(i, j) + \lfloor \gamma_q(x_{H \times (i-1) + j}, y_{H \times (i-1) + j}) \rfloor n) \pmod{256}$$

$$= (I(i, j) + \lfloor \sqrt[n]{(x_{H \times (i-1) + j})^n + (y_{H \times (i-1) + j})^n} \rfloor) \pmod{256},$$

where $I(i, j)$, $I_M(i, j)$ and $r(p, q, a, b, n)_{H \times (i-1) + j}$ is the i -th j -th and $(H \times (i-1) + j)$ -th element of I , I_M and $\mathcal{R}(p, a, b, q, n)$, respectively, and $\gamma_q(x_{H \times (i-1) + j}, y_{H \times (i-1) + j}) = (x_{H \times (i-1) + j})^q \cdot (y_{H \times (i-1) + j})^q$.

The proposed masking phase is applied on different images, binary and gray scale, of different dimensions. Some of the diffused images based on the proposed masking scheme with different masking parameters are shown in Fig. 2. It is evident from the experimental results that the proposed masking scheme is capable of creating high diffusion in the plain-image.

7.1.2. The confusion generation phase

Step 1. Calculate ΣI_M .

Step 2. Select another prime p' satisfying the expression (5).

Step 3. Select two non-negative real numbers k_3 and k_4 , and calculate the parameters a' and b' of OEC for the generation of dynamic S-box using

$$a' = \left(k_3 + \sum I_M\right) \pmod{p'}, \quad (18)$$

$$b' = \left(k_4 + \sum I_M\right) \pmod{p'}. \quad (19)$$

Step 4. Generate the EC $E'(p', a', b')$ and select another natural number ℓ satisfying the expression (6).

Step 5. Generate the S-box $S_{p', \ell}^{a', b'}$ based on the method proposed in Section 4.1.1. Notice that we might not get an S-box for all values of p' , a' , b' and ℓ . For this reason it is better to choose a large prime p' and a small value of ℓ so that the S-box generation probability becomes high.

Step 6. Finally, generate the encrypted image I_C by creating confusion in the pixels of I_M by applying $S_{p', \ell}^{a', b'}$ on each of its pixels.

The proposed encryption technique is applied to images with different properties such as dimension, intensity levels and range of encryption parameters. Some of the encrypted images are shown in Fig. 2.

7.2. Decryption of the proposed cryptosystem

The decryption process is done in the reverse order to the proposed encryption process. That is, first calculate the mask image I_M from encrypted image I_C by using the inverse of the S-box used in the encryption process. Now, to generate the inverse $inv(S_{p', \ell}^{a', b'})$ of the S-box $S_{p', \ell}^{a', b'}$ it is necessary to send the parameters p' , a' , b' and ℓ of S-box generation method as keys to the receiver. If we assume that the channel is noiseless then the application of $inv(S_{p', \ell}^{a', b'})$ on I_C results in I_M . Now, in the second step of the decryption process, the plain-image I is obtained from the I_M . For this step, it is necessary to generate the random number set $\mathcal{R}(p, a, b, q, n)$, which can be obtained by sending the parameters p , a , b , q and n of the proposed PRN generation method as keys. Now, I is obtained by solving Equation (20) for each pixel $I(i, j)$ of I

$$I(i, j) = (I_M(i, j) - r(p, q, a, b, n)_{H \times (i-1) + j}) \pmod{256}. \quad (20)$$

7.3. Security performance

The security strength of the proposed encryption system is analyzed in this section by performing some of the well-known security tests. We used images of Circuit $_{450 \times 600}$, Boat $_{256 \times 256}$, Lena $_{256 \times 256}$ and Pepper $_{512 \times 512}$ for experiments.

7.3.1. Lossless encryption

From the decryption process it is evident that the proposed security system is lossless.

7.3.2. Differential attack

An encryption technique is secure against differential attack if it can generate significantly different cipher images of two slightly different plain images. Different methods are used to quantify the resistance of an encryption technique against differential attack. The number of pixel change rate (NPCR), and the unified average changing intensity (UACI) are well-known efficient methods to measure this resistance. In these tests, the encrypted images I_C and I'_C of two plain-images $I_{N \times M}$ and $I'_{N \times M}$, respectively, which differ at only one pixel are used. Let $D_{N \times M}$ be the binary matrix whose entries are obtained by

$$D(i, j) = \begin{cases} 1 & \text{if } I_C(i, j) - I'_C(i, j) \neq 0 \\ 0 & \text{otherwise} \end{cases}, \quad (21)$$

where $I_C(i, j)$, $I'_C(i, j)$ and $D(i, j)$ are the i -th j -th entries of I_C , I'_C and D , respectively. Then the NPCR and the UACI of a security system are calculated by

$$NPCR = \sum_{i,j} \frac{D(i, j)}{N \times M} \times 100\%, \quad (22)$$

$$UACI = \sum_{i,j} \frac{|I_C(i, j) - I'_C(i, j)|}{N \times M \times 255} \times 100\%. \quad (23)$$

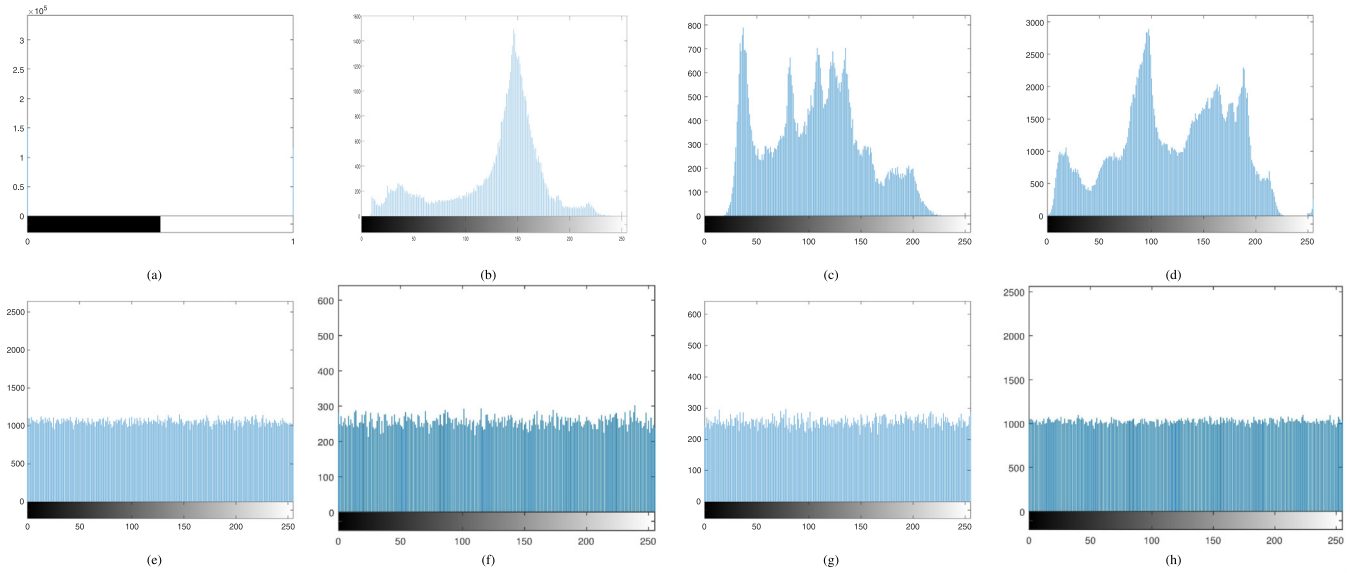


Fig. 3. (a-d) Histogram of plain-images (a-d) in Fig. 2 and (e-h) Histogram of cipher images (i-l) in Fig. 2

Table 8

NPCR and UACI results of the proposed encryption scheme.

Image	Circuit	Boat	Lena	Pepper
NPCR	99.5796	99.5956	99.5964	99.6117
UACI	33.6686	33.2872	33.4762	33.5106

A security system has good resistance against differential attacks if its NPCR and UACI are in the intervals [99,100] and [33,34], respectively.

We implemented the NPCR and the UACI test on many images of different sizes by varying the value of a single pixel at different positions. Some of the experimental results are presented in Table 8. Notice that the NPCR and UACI results lie in the recommended range. Hence the proposed encryption system is secure against differential attacks.

7.3.3. Statistical analysis

It is necessary for a secure cryptosystem to efficiently resist any type of statistical attacks. Thus, we applied several rigorous tests including histogram, correlation and entropy tests to check the performance of the proposed cryptosystem against statistical attacks.

1. Histogram test: A cryptosystem can efficiently resist statistical attacks if the histogram of the cipher image is uniformly distributed. We implemented this test on many images and results of some of the images are shown in Fig. 3.

It is clear from Fig. 3 that the histogram of images after encryption by the proposed technique is uniformly distributed and hence the proposed security system satisfies the histogram test.

2. Correlation test: The pixels of a plain-image are highly correlated with the adjacent pixels so that they can make some meaningful shape. For a secure cryptosystem it is necessary to produce cipher images with very low correlation among the pixels so that the cryptanalyst cannot get any information. We applied the correlation test by calculating the correlation coefficient in the horizontal, diagonal and vertical directions of the adjacent pixel using

$$\sigma_{xy} = \frac{cov(x, y)}{\sqrt{K(x)}\sqrt{K(y)}}, \quad (24)$$

where

$$cov(x, y) = \frac{1}{M} \sum_{i=1}^M (x_i - E(x))(y_i - E(y)), K(x) = \frac{1}{M} \sum_{i=1}^M (x_i - E(x))^2,$$

$$E(x) = \frac{1}{M} \sum_{i=1}^M x_i \text{ and } M \text{ is the data size.}$$

The experimental results of the correlation test are presented in Table 9. It is evident from the results that the correlation among the adjacent pixels of the cipher-image is very close to zero and hence the proposed encryption system satisfies the correlation test.

3. Image entropy test: The entropy test is applied by using Equation (12) to measure the extent of randomness produced by the proposed encryption system in an image of 8-bits and the results are presented in Table 10. Note that the entropy of the cipher images is approximately equal to the optimal value 8 and hence the proposed cryptosystem creates high randomness in the cipher image.

7.3.4. Key analysis

Key attacks are commonly used by the cryptanalyst to break the security of a cryptosystem. Thus for a secure security system, it is important to have high key strength. Key spacing, key sensitivity and known plain-text and chosen plain-text tests are commonly used for key analysis. A brief description of these tests and experimental results of the proposed cryptosystem are given below.

1. Key spacing analysis: The resistance of a security technique against a brute-force attack is measured by its key spacing. In general, a cryptosystem satisfies the key space analysis if it has key spacing more than 2^{128} . In the proposed security technique, nine keys $p, a, b, q, n, p', a', b'$ and ℓ are used as explained in decryption section. We represent each of these keys by 512-bits. This implies that the key spacing of the proposed algorithm is much larger than 2^{128} . Hence, the proposed encryption technique resists brute-force attacks efficiently.
2. Key sensitivity analysis: A secure cryptosystem should be highly sensitive to the keys. This means that a slight change in the keys results in a significantly different cipher-image of the same plain-image. The proposed cryptosystem uses random numbers and a dynamic S-box generated by ordered elliptic curve (OEC). Equations (15), (16), (18) and (19) show dependence of the parameters of OECs used in the proposed cryptosystem on

Table 9
Correlation test results of the proposed encryption scheme.

Image	Correlation of plain-image			Correlation of cipher-image		
	Horizontal	Diagonal	Vertical	Horizontal	Diagonal	Vertical
Circuit	0.9765	0.9550	0.9782	-0.0007	-0.00005	0.003
Boat	0.8834	0.9452	7.1587	0.0005	-0.0011	0.0009
Lena	0.9749	0.9628	0.9864	0.0012	0.0003	0.0010
Pepper	0.9749	0.9640	0.9792	0.0012	-0.0015	-0.0017

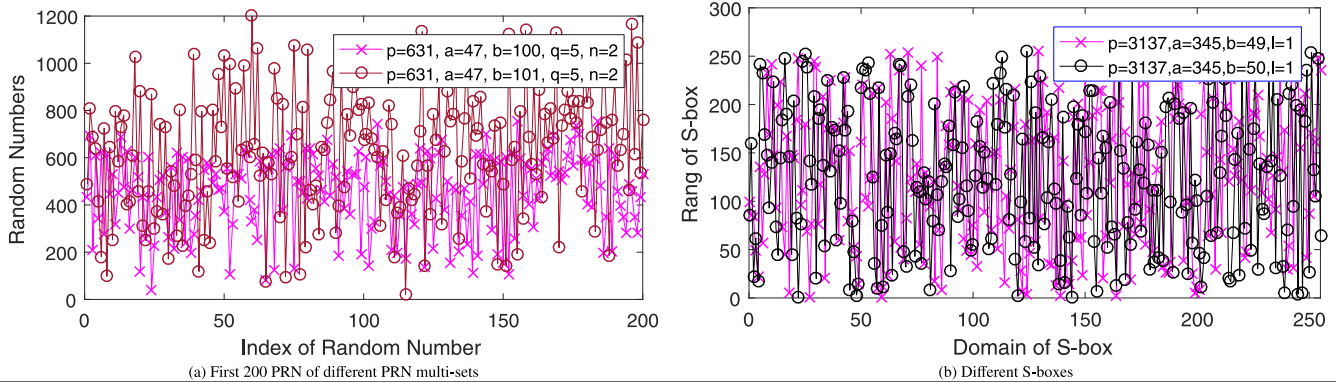


Fig. 4. Comparison of different random number multi-sets and S-boxes generated by the proposed algorithms.

Table 10
Entropy test results of the proposed encryption scheme.

Image	Circuit	Boat	Lena	Pepper
Entropy of plain-image	1.3904	7.1587	7.3838	7.5937
Entropy of cipher image	7.9796	7.9973	7.9993	7.9994

the plain-image. Moreover, OECs are highly sensitive to their parameters, and hence the random numbers and dynamic S-box generated by the proposed methods are highly sensitive to the keys. The sensitivity of the random numbers and dynamic S-box is shown in Fig. 4 by generating the random number sets $\mathcal{R}(631, 47, 100, 5, 2)$ and $\mathcal{R}(631, 47, 101, 5, 2)$, and S-boxes $S_{3137,1}^{345,49}$ and $S_{3137,1}^{345,50}$ by slightly changing the parameters. From Fig. 4, it is easy to observe a significant difference among the random numbers and S-boxes generated by the proposed methods with a small difference in parameters in both random sets and S-boxes. This ultimately implies that the proposed encryption system satisfies the key sensitivity analysis.

- Known plain-text and chosen plain-text attack: In this attack, a ciphered-image is decrypted by using the security keys of some other image. Generally, the cryptanalyst uses all black, or all white images to understand the encryption patterns [30]. Therefore, a security system will have high resistance against known plain-text and chosen plain-text attack if it can efficiently creates diffusion and confusion in all black, and all white images. The results of the proposed security system on these images of size 256×256 are shown in Fig. 5 and Table 11. It is evident from the experimental results that the proposed cryptosystem is secure against known plain-text and chosen plain-text attack.

7.4. Comparison and discussion

A comparison of the proposed encryption system and some of the existing cryptosystems based on different mathematical structures including EC and chaos presented in [30,42–45,48–51] is conducted in this section. All security tests are applied on the listed

security system by using the Lena image of size 256×256 . The experimental results are presented in Table 12.

It is evident from Table 12 that the security results of the proposed encryption technique are comparable with the different type of encryption algorithms. The technique in [30] uses a PRN generated by the group law on an EC while the proposed technique uses a search method to generate random numbers on an OEC. Hence the random number scheme in [30] takes more time than the proposed scheme, since the group law method involves more operations than the search method. Furthermore, the security system in [30] uses AES as a static S-box which makes this scheme vulnerable against algebraic attacks and data analysis attacks as discussed in Sections 3 and 6. However, the proposed encryption system uses a single dynamic S-box generated by the points on an OEC which has high algebraic complexity as shown in Table 2. Hence, the proposed S-box is more secure than the technique in [30]. The scheme in [48] uses random numbers generated by chaotic maps. Thus, the proposed scheme is efficient than the scheme in [48], since the random numbers generated by an EC are more secure than the PRN generated by chaotic maps [36]. In a similar way to [30], the cryptosystem in [48] uses one static S-box for encryption of all type of images and requires multiple rounds to achieve the standard security level. But the proposed scheme uses dynamic S-box whose generation is linked with the plain-image by using Equations (15), (16), (18) and (19), and satisfies all security tests in a single round. This makes the proposed algorithm fast and more robust than [48]. Similarly, the techniques presented in [44,45,49,50] are based on random numbers and multiple S-boxes. Since, the computation increases with the increase in the number of S-boxes. Therefore, the proposed technique has less time complexity than the techniques presented in [44,45,49,50] and provides high resistance against cryptanalysis. The techniques in [42,43,51] are using a combination of EC and chaotic maps at different stages of the encryption process instead of using an S-box. From Table 2, it can be seen that the proposed encryption technique has better cryptographic properties as compared to the techniques in [42,43,51]. Furthermore, the technique in [43] losses information of the plain-image during decryption process, while the proposed technique is lossless. This implies the

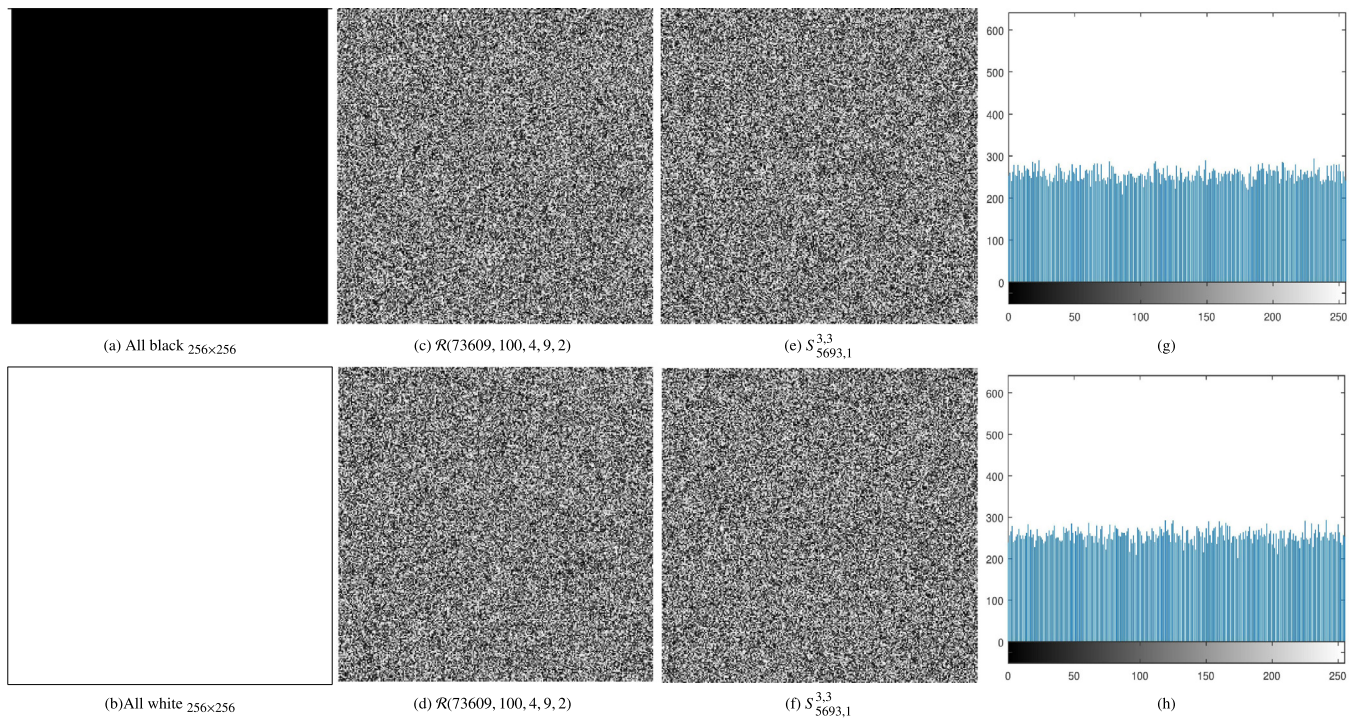


Fig. 5. (a-b) plain-images, (c-d) masked images, (e-f) encrypted images by the proposed method and (g-h) their histograms.

Table 11

Security analysis of all black and all white encrypted images by the proposed encryption system.

Image	NPCR	UACI	Correlation			Entropy
			Horizontal	Diagonal	Vertical	
Encrypted all black	99.6017	33.5034	0.0046	0.0064	0.0051	7.9974
Encrypted all white	99.6460	33.4633	-0.0062	-0.0042	0.0012	7.9971

Table 12

Comparison of the proposed encryption technique with some of the existing cryptosystem by using encryption of Lena_{256 × 256}.

Algorithm	NPCR	UACI	Correlation			Entropy	# S-boxes	Dynamic S-boxes
			Hori	Diag.	Ver.			
Proposed	99.60	33.50	0.0012	0.0003	0.0010	7.9993	1	Yes
Ref. [30]	99.60	33.48	-0.0004	-0.0018	0.0001	7.9993	1	No
Ref. [48]	99.61	33.38	-	-	-	7.9992	1	No
Ref. [49]	99.62	33.70	0.0094	0.0048	0.0112	7.9963	8	Yes
Ref. [45]	99.61	32.79	0.0097	0.0178	0.0136	7.9971	>1 <50	Yes
Ref. [50]	99.62	33.44	0.0018	0.0011	0.0012	7.9030	>1	Yes
Ref. [44]	99.60	33.47	0.0032	-0.0002	0.0011	7.9993	>1	Yes
Ref. [42]	99.60	33.48	0.0029	0.0127	0.0135	7.997	0	-
Ref. [51]	99.50	33.30	0.0010	0.0017	0.125	7.9973	0	-
Ref. [43]	100	33.47	0.0001	0.0089	0.0091	7.9912	0	-

newly developed technique is better than [42,43,51] in the sense of resistance against cryptographic attacks without losing information about the plain-image.

8. Conclusion

A hybrid image encryption technique based on a dynamic S-box, and pseudo random numbers (PRN) over an ordered elliptic curve (OEC) is proposed in this article. The x -coordinates of the OEC followed by modulo 256 operation are used for the generation of S-boxes, while the concepts of a generalized Frobenius map, and n -norm are used on an OEC for the generation of PRN. The newly developed image cryptosystem first masks the plain-image by us-

ing the proposed PRN, and then creates confusion in the masked image by using a dynamic proposed S-box.

An interesting future problem will be to find a method to encrypt a color image with only a single round of the proposed algorithm without increasing the computation complexity.

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